

```
# ===== LIST PROGRAMS
```

```
=====
```

```
# Q1 Sum of all items
lst = [10, 20, 30, 40]
total = 0
for i in lst:
    total += i
print("Q1 Sum =", total)
```

```
# Q2 Largest number
lst = [10, 50, 30, 90, 20]
largest = lst[0]
for i in lst:
    if i > largest:
        largest = i
print("Q2 Largest =", largest)
```

```
# Q3 Smallest number
lst = [10, 50, 30, 90, 20]
smallest = lst[0]
for i in lst:
    if i < smallest:
        smallest = i
print("Q3 Smallest =", smallest)
```

```
# Q4 Count special strings
```

```
lst = ['abc', 'xyz', 'aba', '1211']
count = 0
for i in lst:
    if len(i) >= 2 and i[0] == i[-1]:
        count += 1
print("Q4 Count =", count)
```

# Q5 Common member in two lists

```
list1 = [1, 2, 3]
list2 = [3, 4, 5]
result = False
for i in list1:
    if i in list2:
        result = True
        break
print("Q5 Common Exists =", result)
```

# Q6 Remove 0th, 4th, 5th elements

```
lst = ['Red', 'Green', 'White', 'Black', 'Pink',
'Yellow']
new_list = []
for i in range(len(lst)):
    if i not in [0, 4, 5]:
        new_list.append(lst[i])
print("Q6 Result =", new_list)
```

# Q7 Remove even numbers

```
lst = [1, 2, 3, 4, 5, 6]
new_list = []
for i in lst:
    if i % 2 != 0:
        new_list.append(i)
print("Q7 Result =", new_list)
```

```
# Q8 Find index
lst = [10, 20, 30, 40]
item = 30
index = -1
for i in range(len(lst)):
    if lst[i] == item:
        index = i
        break
print("Q8 Index =", index)
```

```
# Q9 Frequency of elements
lst = [1, 2, 2, 3, 3, 3]
freq = {}
for i in lst:
    if i in freq:
        freq[i] += 1
    else:
        freq[i] = 1
print("Q9 Frequency =", freq)
```

```
# Q10 Remove all occurrences
lst = [1, 2, 3, 2, 4, 2]
element = 2
new_list = []
for i in lst:
    if i != element:
        new_list.append(i)
print("Q10 Result =", new_list)
```

```
# ===== TUPLE PROGRAMS
=====
```

```
# Q1 Length
t = (1, 2, 3, 4)
count = 0
for i in t:
    count += 1
print("Tuple Q1 Length =", count)
```

```
# Q2 Index
t = (10, 20, 30, 40)
item = 30
index = -1
for i in range(len(t)):
    if t[i] == item:
        index = i
        break
```

```
print("Tuple Q2 Index =", index)
```

```
# Q3 Check exists
```

```
t = (1, 2, 3, 4)
```

```
item = 3
```

```
found = False
```

```
for i in t:
```

```
    if i == item:
```

```
        found = True
```

```
print("Tuple Q3 Exists =", found)
```

```
# Q4 Sum
```

```
t = (1, 2, 3, 4)
```

```
total = 0
```

```
for i in t:
```

```
    total += i
```

```
print("Tuple Q4 Sum =", total)
```

```
# Q5 Count occurrences
```

```
t = (1, 2, 2, 3, 2)
```

```
item = 2
```

```
count = 0
```

```
for i in t:
```

```
    if i == item:
```

```
        count += 1
```

```
print("Tuple Q5 Count =", count)
```

```
# Q6 Multiply elements
```

```
t = (1, 2, 3, 4)
```

```
result = 1
```

```
for i in t:
```

```
    result *= i
```

```
print("Tuple Q6 Multiply =", result)
```

```
# Q7 Operations
```

```
t = (10, 20, 30, 20)
```

```
# Length
```

```
length = 0
```

```
for i in t:
```

```
    length += 1
```

```
print("Tuple Q7 Length =", length)
```

```
# Max
```

```
max_val = t[0]
```

```
for i in t:
```

```
    if i > max_val:
```

```
        max_val = i
```

```
print("Tuple Q7 Max =", max_val)
```

```
# Min
```

```
min_val = t[0]
```

```
for i in t:
```

```
    if i < min_val:
```

```
        min_val = i
```

```
print("Tuple Q7 Min =", min_val)
```

```
# Count
```

```
count = 0
```

```
for i in t:
```

```
    if i == 20:
```

```
        count += 1
```

```
print("Tuple Q7 Count =", count)
```

```
# Index
```

```
index = -1
```

```
for i in range(len(t)):
```

```
    if t[i] == 20:
```

```
        index = i
```

```
        break
```

```
print("Tuple Q7 Index =", index)
```

```
# ===== DICTIONARY  
PROGRAMS =====
```

```
d = {}
```

```
# Add
```

```
d['a'] = 10
```

```
d['b'] = 20
```

```
# Remove
```

```
del d['a']
```

```
# Check key
print("Dict Exists =", 'b' in d)
```

```
# Update
d['b'] = 50
```

```
# View
print("Keys =", d.keys())
print("Values =", d.values())
```

```
# Delete all
d.clear()
```

```
# Iterate
d = {'x': 1, 'y': 2}
for key in d:
    print("Dict Item =", key, d[key])
```

```
# ===== STRING
PROGRAMS =====
```

```
# Q1 Length
s = "hello"
count = 0
for i in s:
    count += 1
print("String Q1 Length =", count)
```



```
# Q2 Frequency
s = "google.com"
freq = {}
for ch in s:
    if ch in freq:
        freq[ch] += 1
    else:
        freq[ch] = 1
print("String Q2 Frequency =", freq)
```

```
# Q3 Replace repeated first char
s = "restart"
first = s[0]
new_str = first

for ch in s[1:]:
    if ch == first:
        new_str += '$'
    else:
        new_str += ch

print("String Q3 =", new_str)
```

```
# Q4 Swap first & last
s = "hello"
new_str = s[-1] + s[1:-1] + s[0]
```

```
print("String Q4 =", new_str)
```

```
# Q5 Remove odd index chars
```

```
s = "abcdef"
```

```
new_str = ""
```

```
for i in range(len(s)):
```

```
    if i % 2 == 0:
```

```
        new_str += s[i]
```

```
print("String Q5 =", new_str)
```

```
# Q6 Reverse string
```

```
s = "hello"
```

```
rev = ""
```

```
for ch in s:
```

```
    rev = ch + rev
```

```
print("String Q6 Reverse =", rev)
```

```
# Q7 Count vowels
```

```
s = "hello world"
```

```
vowels = "aeiouAEIOU"
```

```
count = 0
```

```
for ch in s:
```

```
    if ch in vowels:
```

```
        count += 1
```

```
print("String Q7 Vowels =", count)
```

# Q8 Count uppercase, lowercase, digits, special

s = "Hello123@#"

upper = lower = digit = special = 0

for ch in s:

if ch.isupper():

upper += 1

elif ch.islower():

lower += 1

elif ch.isdigit():

digit += 1

else:

special += 1

print("Upper =", upper)

print("Lower =", lower)

print("Digits =", digit)

print("Special =", special)